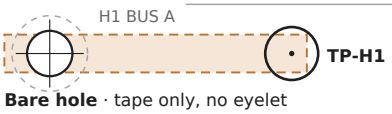
T0 Contact tests

Which joints, part contacts and binder contacts stay reliable? Build on 160–200 gsm card.



Bare hole · tape only, no eyelet

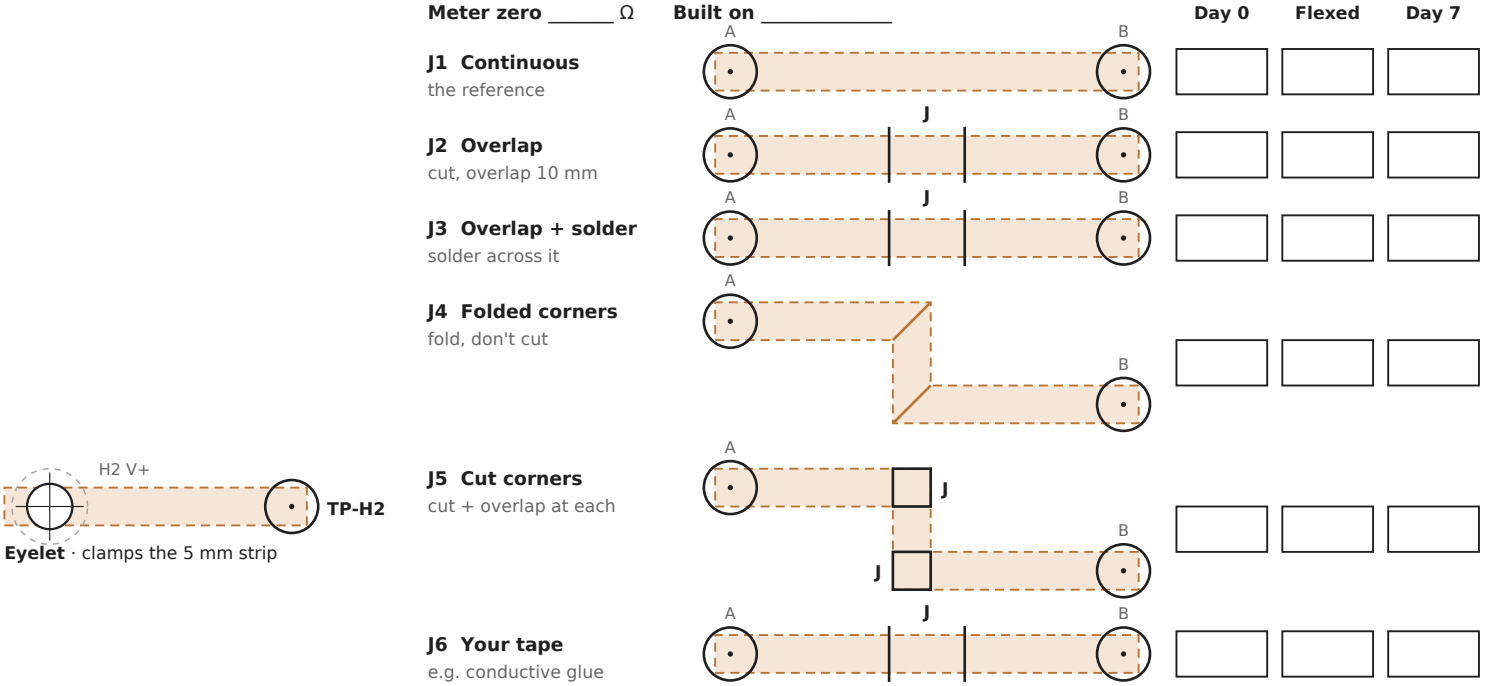
1 Tape joints

Lay each lane from pad A to pad B as marked, then measure A to B in ohms and subtract the meter zero. Write OL for an open circuit. Sheet 2 explains Flexed and Day 7.

Meter zero _____ Ω **Built on** _____

		Day 0	Flexed	Day 7
J1 Continuous	the reference			
J2 Overlap	cut, overlap 10 mm			
J3 Overlap + solder	solder across it			
J4 Folded corners	fold, don't cut			
J5 Cut corners	cut + overlap at each			
J6 Your tape	e.g. conductive glue			

Eyelet · clamps the 5 mm strip



2 Part joints

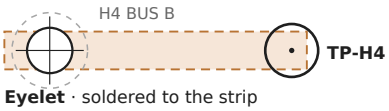
Bridge each gap with a 10 Ω resistor, leads flat on the tape, attached as labelled. Measure A to B. Contact resistance is the reading minus R alone.

R alone _____ Ω one resistor, probes on its leads

	M1 Flat leads, clear tape over	**M2** Flat leads, copper patch over	**M3** Spiral leads, copper patch over	**M4** Soldered	**M5** Your method
Day 0					
Flexed					
Day 7					
Flicker?					



Eyelet · on 12 mm tape



Eyelet · soldered to the strip

Notes

Print at 100 %: this bar should measure 100 mm.

T0 Running the contact tests

Print this sheet once on plain paper. Print sheet 1 on 160–200 gsm card, three copies.

H1 BUS A · NC

You need

- 5 mm copper tape, plus about 15 cm of 12 mm tape for H3
- Five 10 Ω resistors per sheet (brown-black-black)
- Plain brass eyelets for a 6 mm hole, and a setting tool. Not painted aluminium.
- A 2- or 4-hole punch with 80 mm pitch
- A ring binder, and four brass Chicago screws (binding posts) long enough for three sheets
- A multimeter with a 200 Ω range
- A soldering iron for J3, M4 and H4; clear tape

Before you start

1. Print sheet 1 at 100 %. Check the footer bar measures 100 mm with a ruler.
2. Lay the four hole lanes (test 3) first, then punch through the crosshairs, through the tape.
3. Set eyelets in H2, H3 and H4. Leave H1 bare. Solder H4's eyelet to its strip.
4. Zero the meter: 200 Ω range, probes touching. Write the reading on sheet 1 and subtract it from every reading.

H2 V+ · NC

1 · Tape joints

1. Lay each lane as one piece of 5 mm tape, except at the marks: J means cut and overlap the two pieces by 10 mm.
2. J3: solder across the overlap. J4: turn each corner by folding along the printed crease. J5: cut at each corner and overlap.
3. Measure from pad A to pad B. Write OL if the meter shows an open circuit.
4. **Flexed:** bend the card gently over a table edge 10 times, rub each joint firmly with a thumb 5 times, then measure again.
5. **Day 7:** leave the sheet in the binder for a week, then measure again.

2 · Part joints

1. Measure one resistor on its own and write it as R alone.
2. Bridge each station's gap with a resistor, leads flat on the tape. M1: clear tape over flat leads. M2: a copper tape patch over flat leads. M3: coil each lead into a flat spiral, then a copper patch. M4: solder each lead. M5: your own idea.
3. Measure A to B. Contact resistance is the reading minus R alone.
4. **Flicker:** press on the part while watching the meter. Write Y if the reading jumps.
5. Repeat after flexing and on day 7.

10 Ω is low on purpose: a contact of a few ohms shows up clearly against it.

H3 GND · NC

3 · Binder contacts

1. Build the hole treatments on all three copies, as labelled on sheet 1.
2. Bind the three sheets on rings. On each sheet, measure from TP-H*n* to the bare metal of ring *n*. Write the worst of the three readings below.
3. **Turning:** keep the probes on and turn the other sheets back and forth 5 times. Write the highest reading, or OL.
4. Repeat with Chicago screws, finger-tight.

	H1 bare	H2 eyelet	H3 12 mm	H4 solder
Rings, still				
Rings, turning				
Screws, still				
Screws, turning				

Pass or fail

- **Pass:** within 1 Ω of the reference (J1, R alone, or 0 Ω for test 3) and no flicker.
- **Fail:** OL, any flicker, or a reading that drifts by day 7.

Why 1 Ω: lesson circuits use resistors of 100 Ω and up, so 1 Ω changes the current by 1 % or less. Flicker matters more than the exact value.

Results for standard v0.2

Tape overlaps OK?	
Lead attachment	
Hole treatment	
Binding	

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.